Measured on 500 locked test texts

PII redaction with a fine-tuned open model

GLM 5.3 Flash on Fireworks: 84.5% → 92.7% typed F1 after a LoRA fine-tune. Best prompt harness: 88.1%.

Fireworks Deploy Buildathon · Nov 2, 2026 · github.com/renantrendt/fireworks-anonymize

Law firms can’t send client files to an AI service

The rule

Contracts and emails hold names, ID numbers, addresses and phone numbers. Privacy law (LGPD, GDPR) limits who may see them.

The need

Replace every personal value with a fake one before anyone else reads the file.

The gap

Sending raw text to a frontier model exposes it. A stock open model gets the value and label right only 84.5% of the time.

How it works: a router in front of any AI model

1. You ask
in Claude Code or OpenCode, with a contract attached

→

2. Router
a hook catches the text before it leaves

→

3. Company model
the LoRA student on Fireworks lists the personal values

→

4. Mask
each value becomes a fake; the map stays with the router

→

5. Your model
Claude, GPT or any other sees only the fakes

← The answer comes back with fakes. The router swaps the real values back in before you see it.

Today’s benchmark measures step 3. Steps 2, 4 and the return path come from our offline anonymizer.

The task: list every personal value

```
Applicant: Tiago Kruger
ID card: LJ92699VE
IP: 59.227.188.153
Meeting on Sept 12, 11 am
```

→

```
[{"value":"Tiago","label":"GIVENNAME"},
{"value":"Kruger","label":"LASTNAME"},
{"value":"LJ92699VE","label":"IDCARD"},
{"value":"59.227.188.153","label":"IP"}, …]
```

A row passes when the model finds every value and flags no normal word. 25 labels.

Test set and scoring, locked before any run

500

test texts, 3,309 values

Validation split of ai4privacy/pii-masking-300k. 51 texts have no PII, so over-redaction counts. SHA-256 lock.

Typed F1 (headline)

The value must be exact and the label right.

Leak recall (privacy)

Share of values removed from the masked text, whatever the label.

1,000

training rows

Train split only. Same 500 texts, temperature 0, one saved row per text for every system.

Synthetic English text under 512 characters. Legal documents not tested yet.

Baseline: a plain prompt finds the values, mislabels some

17.7%

typed F1, rules only (regex). Names and ID numbers have no fixed format.

84.5%

typed F1, plain prompt on GLM 5.3 Flash (L1)

98.8%

leak recall: almost every value is masked

The errors are wrong labels and boundaries, mostly on ID numbers: ID card 69.1%, driver’s license 70.3%, passport 74.8% F1.

Prompt layers add 3.6 points, then stop

Layer	Change	Typed F1	Leak recall	Sec / text
L0	Rules only (regex)	17.7%	9.8%	0.0
L1	Plain prompt, GLM 5.3 Flash	84.5%	98.8%	4.0
L2	+ label guide	86.5%	97.9%	5.5
L3	+ 3 worked examples	87.7%	98.4%	2.8
L4	+ grounding and regex	87.8%	98.6%	3.1
L5	+ second pass (best harness)	88.1%	99.0%	3.8

Real gains: label guide and examples (McNemar and bootstrap agree). Left unfixed: passport, social number and driver’s license, at 68–72% F1.

Fine-tune: teach the labels prompting could not fix

LoRA on GLM 5.3 Flash

1,000 train-split rows with their correct labels. Rank 8, 3 epochs, 17 minutes on Fireworks.

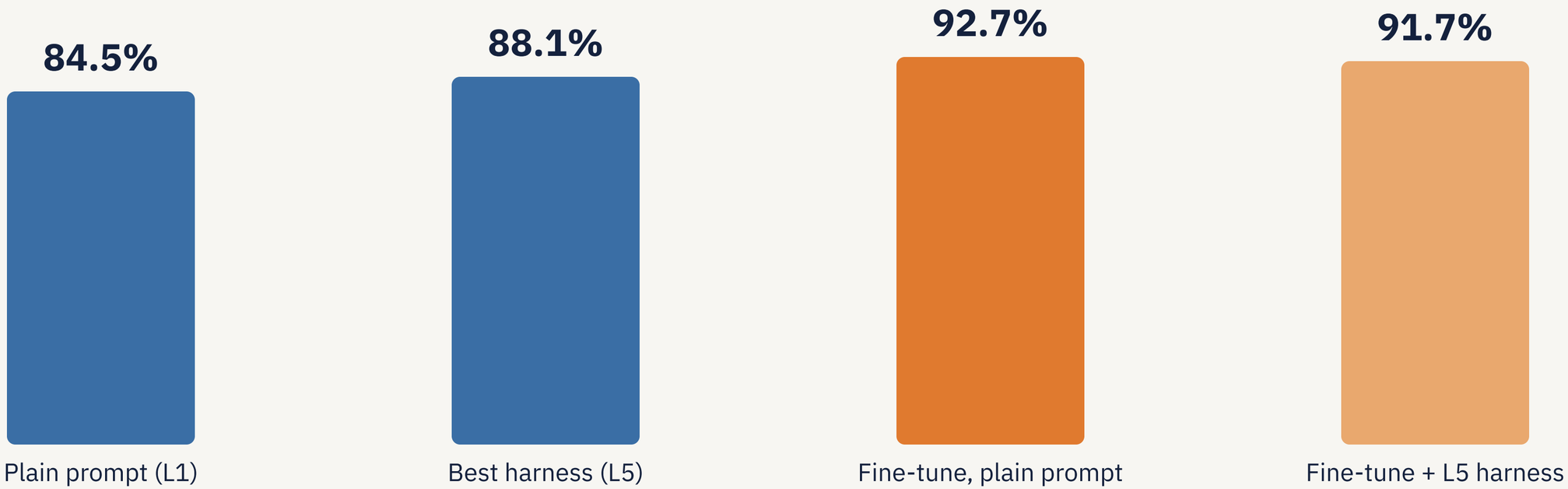
Import set, built

Kimi K3 labeled the same texts; we kept the 567 answers that matched the labels. Not trained yet.

Same harness

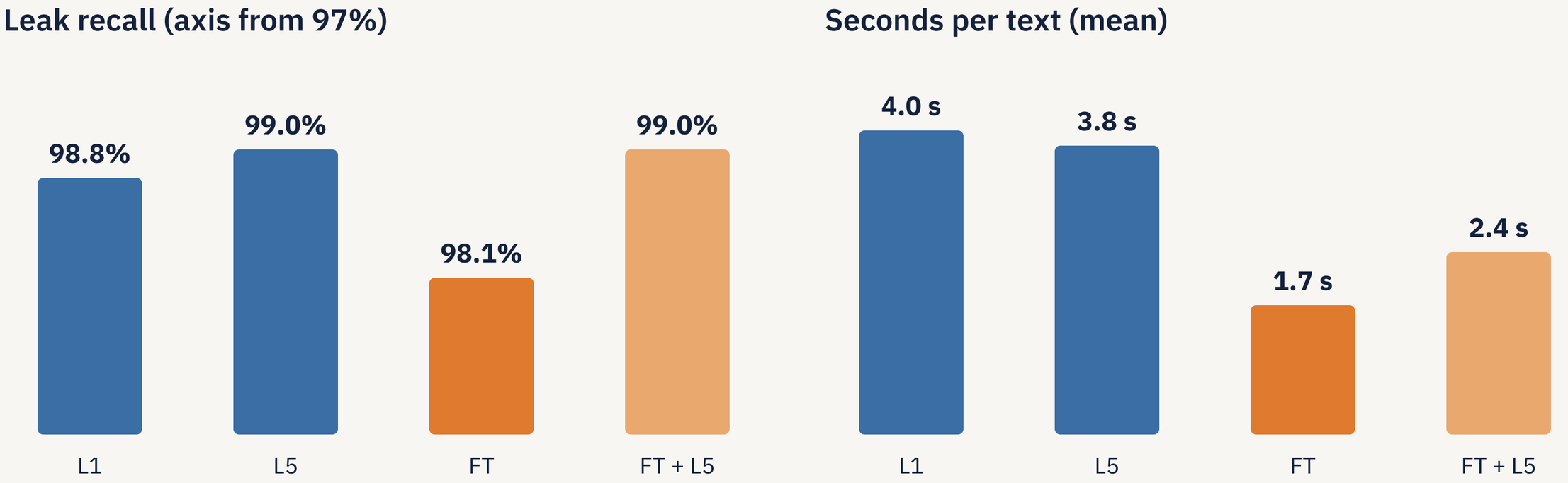
The fine-tuned model runs through the same harness and scorer, with the plain prompt and with full L5.

Results: 92.7% typed F1, +4.6 over the best harness



Typed F1, harness scorer, 500 texts. Fine-tune + L5: 99.0% leak recall and 94.8% of texts with no leak, the version to ship.

Privacy and speed: the fine-tune + harness masks as much, faster



L1 plain prompt, L5 best harness (stock model); FT fine-tune. Texts with no leak: 94.0%, 94.2%, 90.0%, 94.8%. Mean of the final call per text; with rate-limit retries the fine-tune rows are 2.4 and 3.0 s.

Cost and time: $11 and 17 minutes to train

Train

17 min

$11.12, Fireworks job estimate. LoRA, 1,000 rows, 3 epochs.

Test

~3 min

All 500 test texts, 8 in parallel.

Inference

0.9 s

per text (p50). Stock model with 8 examples: 4.0 s.

Tokens per 1,000 texts	Input	Output	Typed F1
Stock, plain prompt (L1)	256k	555k	84.5%
Fine-tune, plain prompt	256k	119k	92.7%
Fine-tune + L5 harness	3,838k	125k	91.7%

Is it real? Two tests, both agree

Comparison	Only before	Only after	McNemar p	Bootstrap 95%
Rows fully correct: fine-tune vs stock	29	160	< 0.0001	+21.4 to +31.0
Rows fully correct: fine-tune vs 8-example prompt	30	136	< 0.0001	+16.6 to +25.8
Values: best harness L5 vs plain L1	48	177	< 0.0001	+2.3 to +4.9

Rows fully correct (value-only scorer): stock 52.8%, 8 examples 57.8%, Kimi K3 56.2%, fine-tune 79.0%. Fine-tune: 0.9 s per text, valid JSON every time.

Against the rubric: strong benchmark, thin business case

Criterion	Weight	What we show	Where we lose points
Benchmark	40%	Rules → prompt → harness → fine-tune on 500 locked texts: 84.5% → 92.7% typed F1; McNemar and bootstrap	Synthetic English data; fine-tune vs best harness not significance-tested
Business value	25%	Legal and compliance teams; live router demo	No user interviews; no cost per document yet
Technical execution	20%	Layered harness, each layer tested; fine-tune; teacher-filtered set	No sweep; RL not run; full pipeline not scored
Presentation & demo	10%	Before/after chart; router demo	Video under 2 minutes
Documentation	5%	One README with every number traced to its file	Training cost not checked against billing

Limitations and next

Limitations

Synthetic English texts under 512 characters.

Fireworks receives raw text; only the company’s own model sees it.

The router is a script, not yet a Claude Code hook.

`github.com/renantrendt/fireworks-anonymize`

Next

1. Train on the teacher set, then RL with the scorer as reward.
2. GPT and Claude on the same 500 texts, with cost per 1,000.
3. Portuguese legal documents.

github.com/renantrendt/fireworks-anonymize